

Vaibhav Suresh Parekh

+1-412-626-1062 | vsparekh@andrew.cmu.edu | [vaibhavparekh9.github.io](https://github.com/vaibhavparekh9) | linkedin.com/in/vaibhavparekh9

EDUCATION

Carnegie Mellon University

Master of Science – Robotics, Mechanical Engineering (Research) | GPA: 3.84/4.00

Pittsburgh, PA

May 2026

University of Mumbai (K. J. Somaiya College of Engineering)

Bachelor of Technology – Mechanical Engineering | GPA: 9.14/10.00

Mumbai, India

May 2022

SKILLS

Tools: Linux, Ubuntu, ROS, ROS2, Gazebo, Nvidia Isaac Sim, Webots, RViz, Foxglove, Git, Docker, SolidWorks, ANSYS

Languages & libraries: C++, Python, MATLAB, Simulink, CMake, OpenCV, MediaPipe, YOLO, PyTorch, CUDA

WORK EXPERIENCE

Lightspeed Robotics

Northbrook, IL

Software and Robotics Engineering Intern

Sep 2025 - Dec 2025

- Built an **LLM-based parser** for BTLx (XML schema for timber fabrication) to translate design data into robotic panel-cutting instructions, enabling AI-driven construction automation.
- Curated datasets and developed the **perception stack** for BRUTE robot (a stationary dual-arm cell for panel construction) for detecting processed lumber at the cutting station.
- Developed **ROS2** based navigation (**Nav2**) and manipulation (**MoveIt**) stacks for DEX robot (a mobile dual-arm manipulator), with simulations in **Nvidia Isaac Sim**.

Inductive Robotics

Austin, TX

Robotics Engineer Intern

Jul 2025 - Aug 2025

- Developed **ROS2** based perception and navigation stack for docking a Clearpath Jackal-based robot to an inductive charger, and simulated the same in **Nvidia Isaac Sim**.
- Generated **synthetic dataset** in Isaac Sim using Replicator and trained a **YOLOv11** model for preemptive detection of reversing vehicles through taillight-state classification, achieving **92% accuracy** on real-world data through effective **sim-to-real** transfer using domain randomization.
- Modeled the robot in **SolidWorks** and authored its **URDF**, along with the simulation environment and assets in Blender.

Tata Power Renewable Energy Limited

Mumbai, India

Lead Associate

Jun 2022 - Aug 2024

- Performed financial and energy modeling using Plexos and Excel to integrate over **6 GW** of energy from wind, solar, and ESS for optimizing power plant capacities to meet diverse power requirements.
- Led a cross-functional project to customize and **fine-tune a GPT-3.5 model** for summarizing, assessing risks, and comparing lengthy tender documents, **reducing TAT** for tender analysis from hours to minutes.

ACADEMIC PROJECTS

Graduate Research (Computational Engineering & Robotics Lab)

Oct 2024 - May 2026

- Devised an unsupervised approach to identify keypoints on multi-viewpoint objects from single RGB image using stacked Hourglass CNNs, achieving **PCK@10 accuracy of 82.8% and 69.5%** for intra- and cross-instance correspondences.
- Rigged robot models in Blender for simulation in Nvidia Isaac Sim, enabling visual **SLAM** implementations and testing.

F1TENTH Autonomous Racing

Jan 2026 - May 2026

- Engineered a **LiDAR-based ROS2 navigation stack** (PID Wall Following, Follow-the-Gap, Pure Pursuit, RRT*, MPC) deployed on **Nvidia Jetson** for an autonomous racecar, utilizing SLAM Toolbox for **mapping** and Particle Filter for **localization**.
- Built an online anti-skid system using **sensor fusion** of wheel-encoder and **IMU** velocities to detect slippage and adaptively optimize velocity profile.
- Achieved 1st place among 9 teams with a **44.6% lower lap time** than the overall average.

Self-Docking Autonomous Robot

Jan 2025 - May 2025

- Led perception stack development for a wall-outlet detection pipeline using **Intel RealSense D435** and **YOLOv8n** trained on a custom dataset, exported to **ONNX** and deployed on **Raspberry Pi 5**.
- Implemented **image-based visual servoing** over **ROS2** to align the mecanum base and actuate pin insertion.
- Prototyped **RTAB-Map** visual SLAM and **Nav2** path planning for the same platform in **Gazebo** simulation.

Planning for Multi-UAV Search and Rescue

Oct 2024 - Dec 2024

- Formulated and simulated an **NBV (next-best-view) planner** algorithm in **MATLAB** leveraging LiDAR-based occupancy maps to identify intermediate goal positions, maximizing UAV coverage efficiency in disaster zones.
- Deployed **RRT algorithm** for efficient UAV navigation to intermediate goal points, enabling faster search-and-rescue coverage.

Controller Design for an Autonomous Buggy

Oct 2024 - Dec 2024

- Programed **PID**, **LQR**, and **Kalman** filter-based controllers for a car to accurately follow predefined trajectories in **Webots**.
- Implemented **A*** for path planning, and **EKF SLAM** for localization and controlling the vehicle in GPS-denied environment.